TECHNICAL IMPLEMENTATION COMPANION

Evidence-First TBML Triage System

Data engineering, time-safe analytics, controlled model evaluation, evidence construction, and deployment

A technical account of how official public trade data is transformed into a ranked queue of unusual corridor-product-year patterns for accountable human review.

25,844	2017-2024	Top 50	27×
OFFICIAL ANNUAL OBSERVATIONS	ANALYTICAL PERIOD	RANKED OFFICIAL QUEUE	TEST LIFT AT 50

Document identity

Version: 1.0

Generated: 23 July 2026

Source commit: `774aac6` on `develop`

Analytical release: `official-data-simplified-v1`

Implementation status

IMPLEMENTED offline analytical pipeline

DEPLOYED read-only Streamlit dashboard

FUTURE STATE bank-data integration

Live project: tbml-review-triage.streamlit.app
Repository: github.com/jessybaolin/evidence-based-tbml-corridor-drift

Human-review boundary. This output prioritises an unusual corridor-product pattern for human review. It does not establish money laundering, misinvoicing, or criminal intent.

FRONT MATTER

Document control

This companion is the technical layer beneath the stakeholder project brief and live dashboard. It documents the current repository implementation rather than a proposed target architecture.

Control item	Current record
Purpose	Explain the implemented data, analytical, modelling, evidence, dashboard, deployment, and control design
Intended audience	Data and analytics practitioners; AFC/AML stakeholders; model-risk reviewers; wholesale-banking technology stakeholders
Implementation version	`official-data-simplified-v1`
Source commit	`774aac6` (`develop`)
Analytical period	2017-2024
Trade dataset	CEPII BACI HS17 release `202601` , filtered to three HS6 products
Benchmark dataset	World Bank Commodity Markets annual workbook, 24 family-year rows
Deployment status	Public-data, read-only dashboard hosted on Streamlit Community Cloud
Owner	Jesslyn Guo Baolin
Related document	`stakeholder_project_brief_v2.pdf`

How to read implementation status

IMPLEMENTED
Code or configuration exists in the repository.

GENERATED
An artefact is produced by a pipeline stage.

VALIDATED
A check or test has executed against the current build.

FUTURE STATE
Design intent only; not part of the current application.

Evidence policy

Every retained metric in this document was checked against current generated outputs or recomputed from current processed data. The working claim-source map is included with the document build at `reports/technical_companion/technical_companion_claim_source_map.csv`.

Original preserved. The seven-page legacy companion remains at `docs/afc_aml_technical_companion.pdf`. This revision is a separate generated artefact.

CONTENTS

Contents: core implementation

Select a section to jump to its technical narrative.

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Contents: outputs, controls, and appendices

Traceability principle
Dashboard statement → source artefact → transformation → code module → interpretation boundary.

Technical executive overview

The project addresses a prioritisation problem. Public trade data contains thousands of annual country-to-country product observations, but no confirmed TBML labels and no customer, invoice, shipment, payment, or ownership records. The implemented system therefore ranks unusual analytical patterns for review; it does not classify financial crime.

25,844 OFFICIAL OBSERVATIONS	23,656 ELIGIBLE FOR SCORING	50 RANKED REVIEW CASES	200 EVIDENCE ROWS

Unit of analysis and output

One row is one exporter-importer-HS6-year aggregate. The selected method scores the 23,656 rows with positive trade value and usable quantity, sorts them deterministically, persists the Top 50, and attaches four evidence checks to each queued observation.

INPUT
Official annual trade
Value, quantity, corridor, product, year

→

ANALYTICS
Time-safe signals
History, peers, benchmark, movement, quality

→

EVALUATION
Controlled scenarios
Used to compare methods, never displayed as cases

→

OUTPUT
Queue plus evidence
Read-only stakeholder dashboard

Offline pipeline, online presentation

The analytical pipeline runs as separate Python stages and writes Parquet, CSV, JSON, joblib, report, and figure artefacts. The Streamlit application loads those generated files, caches them, applies interactive filters, persists the selected case in session state, and renders charts and tables. It does not rebuild the panel, inject scenarios, retrain models, or write data.

> **Design decision: precompute, then present.** This keeps stakeholder interaction fast and makes every displayed number traceable to a versioned artefact. The trade-off is that the dashboard is not a live scoring service.

Scope, intended use, and analytical boundaries

What the system does

- Ranks unusual annual corridor-product-year patterns.
- Compares a row with its prior history, same-year peers, and a broad commodity benchmark.
- Shows data-quality caveats and recomputable evidence.
- Supports an accountable human decision about what to review next.

What it does not do

- Detect, prove, or label money laundering.
- Estimate a probability of criminal activity.
- Determine invoice-level fair value.
- Assign country, customer, or counterparty risk.
- Replace transaction monitoring or investigation.

Boundaries carried through the design

Boundary	Technical consequence
Annual public aggregates	A unit value is an implied annual average, not an invoice price
World Bank benchmark	Used as market context, not corridor-specific fair value
FATF-Egmont publications	Used for typology language and questions, never as features or labels
No confirmed cases	Controlled scenarios evaluate ranking behaviour; they are not ground truth
Missing quantity	Row cannot form unit value and remains outside model scoring
Score semantics	Hybrid score determines ordering only; it is not calibrated
Global SHAP only	Explains aggregate XGBoost contribution, not an individual case

Design decision: retain incomplete rows. Rows that cannot be scored remain in the panel and exclusion audit. Silent deletion would improve headline completeness while hiding a data-quality blind spot.

This output prioritises an unusual corridor-product pattern for human review. It does not establish money laundering, misinvoicing, or criminal intent.

3 · ARCHITECTURE

End-to-end technical lifecycle

01 · OFFICIAL DATA
BACI, World Bank, reference files
Source files are hashed, counted, and schema-checked.

→

02 · PREPARATION
Clean annual panel
Mappings, units, benchmark joins, quality status, canonical IDs.

→

03 · ANALYTICS
Time-safe feature table
Current row, same-year peers, and prior-only corridor history.

↓

04 · EVALUATION
Controlled scenario layer
Rules and models compared using separated scenario labels.

→

05 · OFFICIAL OUTPUT
Top 50 and evidence
Selected scorer returns to official observations only.

→

06 · PRESENTATION
Streamlit dashboard
Cached, read-only access to generated analytical artefacts.

Implemented lifecycle

Verify and ingest	`00_check_environment.py` → `01_verify_sources.py` → `02_ingest_official_sources.py`
Prepare and model	`03_build_panel.py` → `04_build_features.py` → `05_build_scenarios.py` → `06_train_evaluate_models.py`
Explain and report	`07_build_evidence.py` → `08_build_briefs.py` → `09_generate_reports.py`
Present and deploy	`dashboard/streamlit_app.py` reads committed outputs; GitHub-connected Streamlit Cloud starts the app.

Lifecycle boundary
The controlled scenario branch is used to choose and test a ranking method. The official-data branch produces the cases shown to stakeholders.

Figure 1. Technical lifecycle from verified public sources to a deployed, read-only review dashboard.

Data architecture and lineage

Official-data track

Raw	BACI extract; country and HS17 mappings; World Bank workbook
Interim	`raw_baci_selected.parquet`; `worldbank_commodity_benchmarks.parquet`
Processed	`corridor_product_year_panel.parquet`; `corridor_features.parquet`
Outputs	`top_ranked_corridors.csv`; `evidence_table.csv`; deterministic briefs

↔

Evaluation-only track

Test copy	`scenario_panel.parquet` preserves the panel schema
Labels	`scenario_labels.parquet` remains physically separate
Features	`scenario_features.parquet` uses the same feature builder
Evaluation	model scores, comparison, selected parameters, hybrid candidates

Canonical identity

The canonical analytical key is `year + exporter_iso3 + importer_iso3 + hs6`. `obs_id` is a stable hash-derived observation identifier, while `source_row_id` links the prepared row back to the filtered BACI source. Evidence rows add their own `evidence_id`.

Design decision: one feature builder. Official and scenario panels pass through the same `build_features()` function. This reduces train-versus-runtime drift; labels are joined only inside model fitting.

Figure 2. Separate official and evaluation-only data tracks share transformations without sharing scenario labels.

Artefact map: grain, ownership, and consumers

Artefact	Grain / rows	Created by	Main consumer	Purpose
`raw_baci_selected.parquet`	source row / 25,844	stage 02	stage 03	Normalised raw fields and source IDs
`worldbank_commodity_benchmarks.parquet`	family-year / 24	stage 02	stage 03	Annual benchmark context
`corridor_product_year_panel.parquet`	corridor-product-year / 25,844	stage 03	stages 04-05; dashboard	Canonical prepared panel
`exclusion_audit.csv`	caveated or excluded row / 13,950	stage 03	controls; dashboard	Explicit quality and exclusion record
`corridor_features.parquet`	official observation / 25,844	stage 04	stages 06-07; dashboard	Time-safe analytical signals
`scenario_panel.parquet`	copied panel / 25,844	stage 05	stage 06	Controlled transformations only
`scenario_labels.parquet`	observation / 25,844	stage 05	stage 06	Split, scenario ID, positive and hard-negative labels
`model_scores.csv`	eligible scenario row / 23,656	stage 06	evaluation page	Candidate and hybrid scores
`model_comparison.csv`	split-family-model / 40	stage 06	model review; dashboard	Ranking metrics
`top_ranked_corridors.csv`	official queue / 50	stage 06-07	dashboard	Persisted review order
`evidence_table.csv`	evidence check / 200	stage 07	case page; briefs	Recomputable case facts
`analyst_briefs.json`	brief / 5	stage 08	reporting reference	Deterministic, caveated top-case briefs

Required dashboard files fail visibly if absent. Optional files return a labelled empty state; the application never substitutes demo data.

Unit and key controls

- HS6 remains a six-character string.
- BACI `v` is multiplied by 1,000 to obtain current USD.
- BACI `q` is retained in metric tons.
- Gold benchmark USD/troy ounce is converted to USD/metric ton.
- Canonical keys are checked for duplicates.
- Every panel row is retained unless explicitly classified; exclusions remain auditable.

Preparation rules and quality classification

Core transformations

Trade value

$$P@k = \frac{TP_k}{k}$$

BACI reports value in thousands of current USD.

Quantity

$$R@k = \frac{TP_k}{N_+}$$

The selected BACI quantity is already reported in metric tons.

Implied unit value

$$L@k = \frac{P@k}{N_+/N}$$

Calculated only when value and quantity are positive.

Gold benchmark

$$C_{c,p,t} = |\Delta \log UV_{c,p,t} - \Delta \log B_{p,t}|$$

Converts USD/troy ounce to USD/metric ton.

Quality status

`data_quality_score` ranges from 0 to 6: one point for valid quantity, two for a matched benchmark, up to two for history depth, and one for core-field completeness. The status is assigned separately:

- **Fully usable:** valid value and quantity, matched benchmark, enough history, no retained extreme caveat.
- **Usable with caveat:** eligible for modelling but history is short or the row is a valid extreme.
- **Excluded from modelling, retained for audit:** unit value cannot be computed, mainly because quantity is unavailable.

Important distinction. Missing quantity excludes the row. For otherwise eligible rows, lower data quality can reduce the transparent weighted sum by at most 0.08. Data weakness does not increase unusualness.

Official data sources and product-family design

CEPII BACI

Role: annual bilateral trade facts.

Value and quantity by exporter, importer, year, and HS6 product.

Official-derived filtered extract; annual aggregates, not transactions.

World Bank

Role: broad market-price context.

Annual palm-oil, copper, and gold benchmark series.

Not corridor-specific landed price or invoice fair value.

FATF-Egmont

Role: typology language, caveats, and review questions.

Published 2020 trends and 2021 risk-indicator references.

Never a feature, label, or row-level fact.

Why these three HS6 products

Crude palm oil · 151110

High-volume bulk commodity with a usable annual benchmark. It tests behaviour around established physical flows.

Refined copper cathodes · 740311

Industrial metal with a widely followed market reference. It tests benchmark-relative movement.

Non-monetary unwrought gold · 710812

Compact, high-value product relevant to TBML typologies. The code excludes powder and does not reveal purity, form, or contract terms.

The families deliberately vary in volume, value concentration, market structure, and benchmark scale. That variation makes family-level evaluation necessary: a method that works well for gold may not behave the same way for palm oil.

Design decision: one exact HS6 code per family. This sacrifices wider commodity coverage in return for clearer comparisons, stable benchmark mapping, and easier interpretation.

Coverage and economic scale

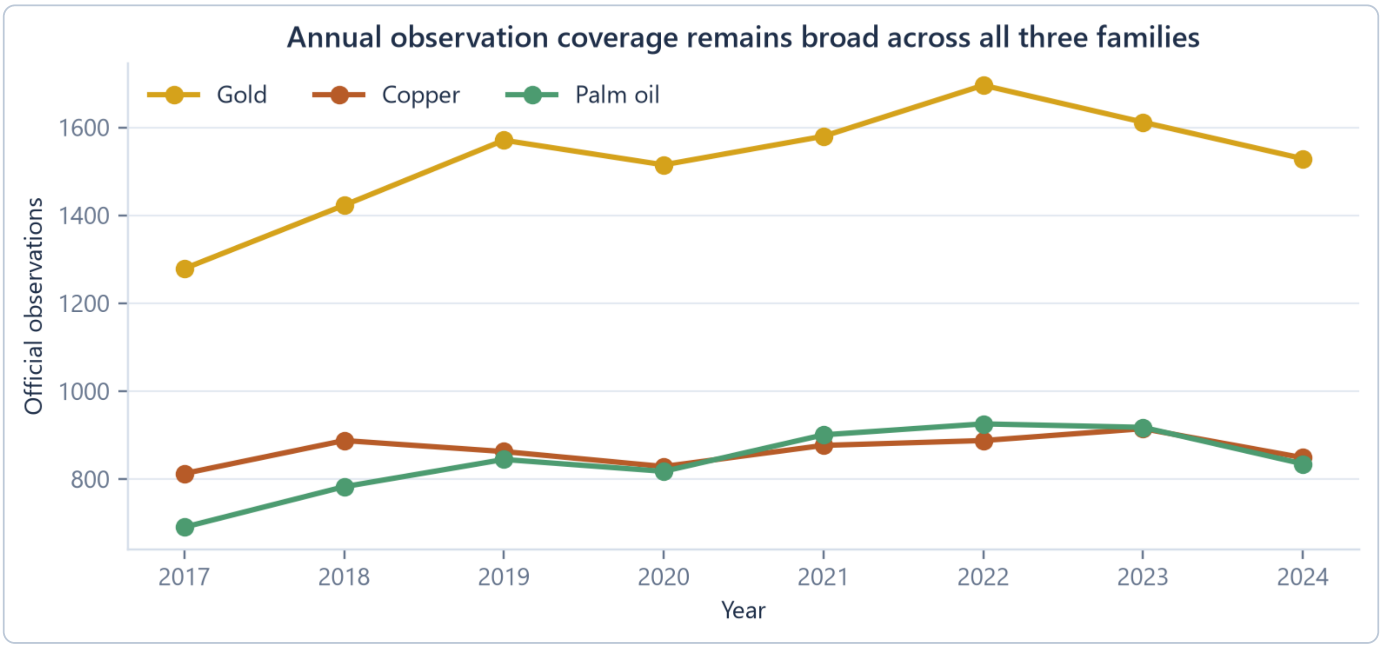

Figure 3. Official row coverage by year and family. Source: current corridor-product-year panel.

The panel contains 25,844 annual observations: 12,205 gold, 6,922 copper, and 6,717 palm-oil rows. Annual coverage ranges from 2,783 rows in 2017 to 3,510 in 2022. The chart describes record availability, not suspiciousness.

Value and quantity tell different stories

Family	Official rows	Trade value	Share of scoped value	Main interpretation
Gold	12,205	USD 2.851tn	81.01%	Value-weighted totals are dominated by gold
Copper	6,922	USD 573.3bn	16.29%	Industrial benchmark context is material
Palm oil	6,717	USD 95.1bn	2.70%	High physical quantity, lower value share

Statistical consequence. A portfolio-level value-weighted statistic is mostly a statement about gold. The dashboard therefore shows each product family separately where scales differ materially.

The profiling notebook also showed large within-family dispersion in implied unit value. That observation motivates log transformations, robust prior-history comparisons, same-year peer positioning, and explicit caveats for small quantities.

From the population to the Top 50

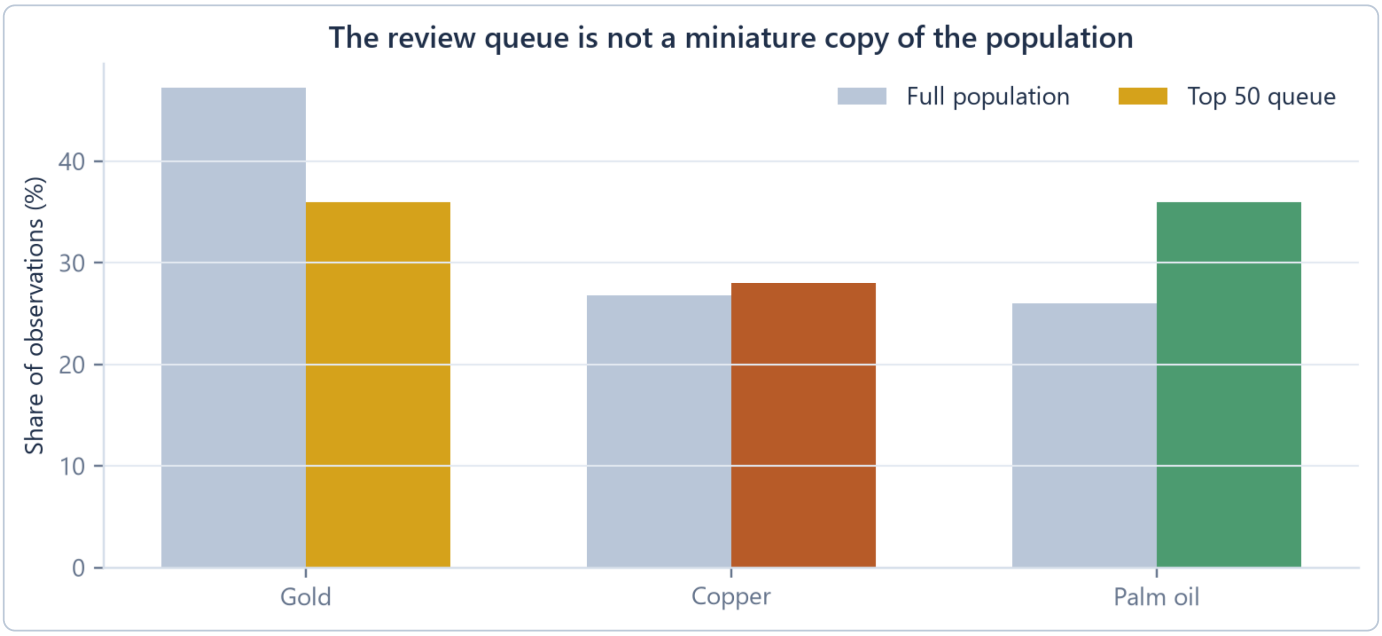

Figure 4. Product-family composition in the full panel and the selected official queue.

The Top 50 contains 18 gold, 18 palm-oil, and 14 copper observations. Palm oil is therefore more prominent in the queue than in the full row population, while gold is less dominant by count than it is by value.

What the queue profile means

- Sixteen of the 50 rows are from 2022, making it the largest queue year.
- Forty-five rows are `usable_with_caveat`; the common reason is a retained valid extreme or limited history, not missing quantity.
- The queued rows together represent only 0.0061% of total scoped trade value. Rank is driven by unusual-pattern signals, not economic size.
- Every queued row has usable value and quantity. Missing quantity cannot create a high unit-value score.

> **Design decision: separate unusualness from materiality.** The current ranker answers “which patterns are analytically unusual?” It does not impose a minimum transaction value. A bank implementation would add customer, transaction, and materiality context before deciding review priority.

The queue is a triage surface, not a prevalence estimate. Its composition must not be interpreted as a claim that one product, year, or country is riskier.

Gold rows without usable quantity

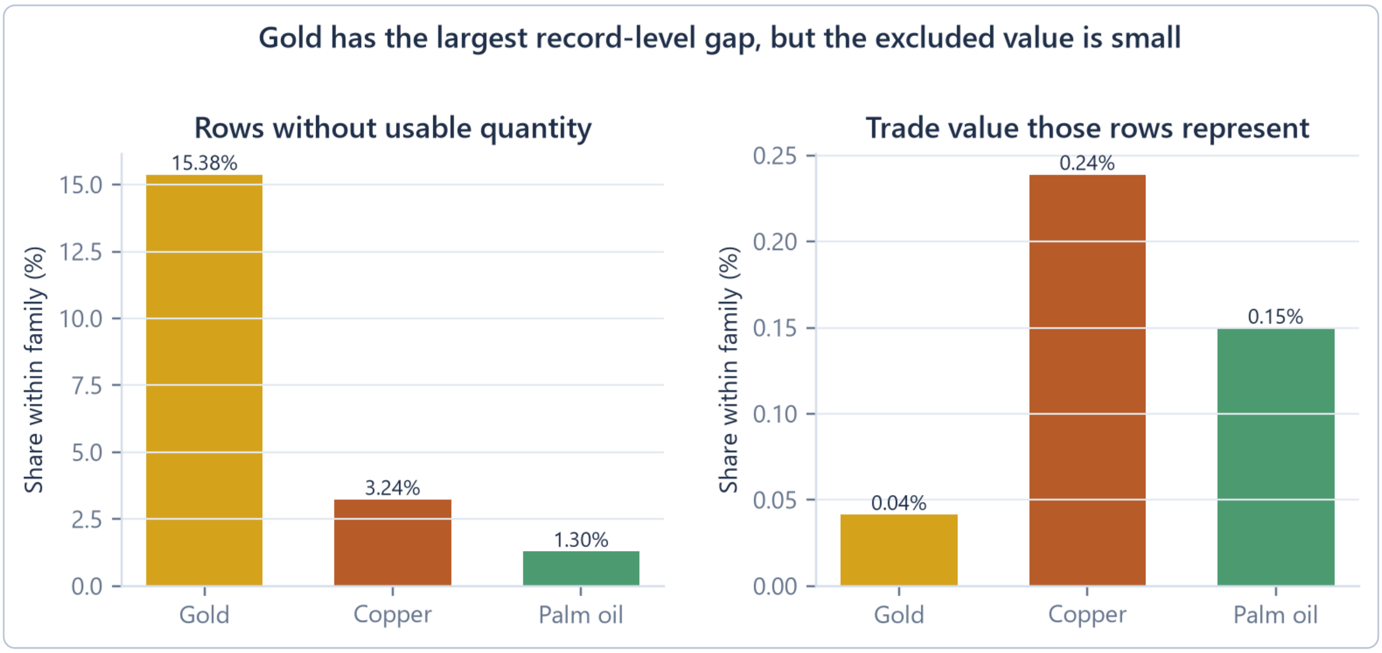

Figure 5. Missing quantity is large by record count in gold but small by reported trade value.

Gold has 1,877 rows without usable quantity, 15.38% of gold records. Those rows represent USD 1.183bn, only 0.04149% of gold value. This is why the analysis is a coverage control rather than an anomaly story: a meaningful share of records cannot be assessed, but almost all reported value remains covered.

What the deeper audit found

One 2024 United Arab Emirates-to-Thailand row accounts for 70.9% of excluded gold value, but quantity is usable in the other seven years of that route. Separately, 14 routes are present in every year from 2017 to 2024 and lack usable quantity in every year. Twelve touch the Netherlands, yet all 14 together carry only USD 1.0m, or 0.08455% of excluded gold value.

Large one-year gap

Financially prominent within the excluded subset; best treated as a source-record check, not a route-wide conclusion.

Small recurring gaps

Operationally traceable because the same routes repeat; not financially material and not a country-risk signal.

Missing quantity is retained for audit, monitored separately, and never converted into review priority.

Feature foundations

The current model input list contains 22 fields. Some are direct or transformed values; others encode history, peers, benchmark movement, corridor status, missingness, and data quality. All are calculated at the corridor-product-year grain.

Level and benchmark features

Log unit value

$$P@k = \frac{TP_k}{k}$$

Compresses the highly skewed unit-value scale. Undefined when value or quantity is non-positive.

Benchmark residual

$$R@k = \frac{TP_k}{N_+}$$

Positive values sit above the annual family benchmark; negative values sit below it.

The residual is a log ratio. A residual of zero means parity with the benchmark; it does not mean “no risk”. The benchmark is a broad market reference and does not observe freight, purity, grade, contract terms, timing, or re-export effects.

Explicit missingness fields

`missing_quantity_flag`, `missing_benchmark_flag`, `missing_history_flag`, and `missing_country_mapping_flag` preserve data limitations for modelling and audit. Model eligibility still requires a valid unit value, so missing quantity rows do not enter fitted or official scores.

Design decision: logs before differences. Log ratios turn multiplicative changes into additive differences and make a doubling comparable in magnitude to a halving. Zero and missing denominators remain excluded rather than silently replaced.

Prior history and same-year peers

Shifted corridor history

For each `corridor_id + hs6`, prior log unit values are accumulated in year order. The current row is excluded before the median and median absolute deviation (MAD) are calculated.

$$P@k = \frac{TP_k}{k}$$

$$R@k = \frac{TP_k}{N_+}$$

$$L@k = \frac{P@k}{N_+/N}$$

The implementation uses the unscaled MAD, not the normal-consistency factor `1.4826`. If prior MAD is zero or smaller than `1e-9`, the denominator becomes `0.05`. If no prior median exists, the robust z-score remains missing. This can produce large values for short, low-dispersion histories, so history depth is always shown as a caveat.

Same-family, same-year peer percentile

$$C_{c,p,t} = |\Delta \log UV_{c,p,t} - \Delta \log B_{p,t}|$$

The percentile ranks the row's benchmark residual among all corridors in the same family and year. Same-year peers are available at assessment time, so this comparison is time-safe even though it is not prior-only.

Trade-off: contemporaneous context. Peer position removes a market-year scale effect, but the peer group can still mix legitimate differences in grade, trade terms, and route structure.

Movement, benchmark drift, and corridor status

For the previous observed row in the same corridor and HS6 group:

$$P@k = \frac{TP_k}{k}$$

This convention produces `trade_value_yoy_change`, `quantity_yoy_change`, and `unit_value_yoy_change`. The "previous" observation is the prior available row, not necessarily the immediately preceding calendar year.

Value-quantity divergence

$$R@k = \frac{TP_k}{N_+}$$

Shows whether value changed faster than reported physical quantity.

Benchmark-adjusted drift

$$L@k = \frac{P@k}{N_+/N}$$

Shows whether the corridor's gap to the market benchmark changed.

Benchmark-consistency gap

$$C_{c,p,t} = |\Delta \log UV_{c,p,t} - \Delta \log B_{p,t}|$$

Near zero means corridor unit value moved similarly to the market.

Corridor status

`corridor_activity_history` counts earlier appearances. `novelty` marks the first. `reactivation` marks a return after a gap of more than one year.

`valid_extreme_flag` records unit value above five times or below 0.2 times the benchmark after basic validity checks. It is a caveat-bearing analytical flag, not proof of mispricing.

Time-safety and feature use

Time-safe means no future information.
When a 2022 row is assessed, its history features may use 2017-2021. Same-year 2022 peers and the 2022 benchmark are allowed. Data from 2023-2024 is unavailable to that row.

2017-2020

Prior observations

May contribute to shifted history for a later row.

→

2021

Most recent prior row

Used for lagged changes when available.

→

2022

Row being assessed

Current facts, 2022 benchmark, and 2022 peers are available.

→

2023-2024

Hidden future

Never enters the 2022 feature vector.

Feature roles

Feature family	Time-safe reference	Weighted sum	ML	Evidence/dashboard
Unit value and logs	current row	indirect	yes	yes
Prior median, MAD, robust z	prior observations only	yes	yes	yes
Same-year peer percentile	same family and year	yes	yes	yes
YoY log changes and divergence	previous observed row	yes	yes	yes
Benchmark residual and drift	current benchmark; prior residual	yes	yes	yes
Novelty and reactivation	prior activity count/year	yes	yes	yes
Missingness and quality	current data state	reduction / eligibility	yes	yes

Figure 6. Feature construction timeline. The current row can use prior history and contemporaneous market context, never future years.

Controlled evaluation scenarios

Public BACI rows have no confirmed TBML outcome. The project therefore evaluates ranking behaviour on a frozen copy of the clean panel with controlled transformations.

Official observations

Unchanged public rows are feature-engineered and scored after model selection.

Output: official Top 50 and evidence

≠

Controlled scenarios

540 selected rows are modified on a test copy: 108 positives and 72 hard negatives in each split.

Output: model-comparison metrics only

Scenario catalogue

Evaluation role	Scenario	Controlled change
Positive	Over-valuation	Value multiplied 2.2-3.5×; quantity held
Positive	Under-valuation	Value multiplied 0.25-0.45×; quantity held
Positive	Low volume, high value	Value rises 1.8-2.8×; quantity falls to 0.35-0.65×
Hard negative	Benchmark-consistent movement	Unit value aligned with the family-year median benchmark residual
Hard negative	Proportional growth	Value and quantity multiplied by the same factor

Scenario IDs and labels live in `scenario_labels.parquet`, not in the scenario panel or feature table. Forbidden label-like fields are explicitly rejected by `build_features()`.

Limitation. Success on planted patterns shows whether the method recovers the behaviours it was designed to recognise. It is not a real-world crime-detection rate.

Transparent weighted-sum score

Nine signals are first normalised to the range 0-1 using fixed reference values. Their documented points are added, then two reductions are subtracted.

$$P@k = \frac{TP_k}{k}$$

Positive component	Weight	Reference / construction
Prior-history deviation	0.20	`abs(robust_z) / 4`, clipped
Benchmark residual	0.18	`abs(residual) / 1`, clipped
Benchmark-adjusted drift	0.18	`abs(drift) / 0.70`, clipped
Unit-value movement	0.12	`abs(unit-value change) / 0.80`, clipped
Value-quantity divergence	0.12	`abs(divergence) / 0.80`, clipped
Peer-tail position	0.10	distance from percentile 0.5, doubled
New corridor	0.06	binary
Reactivated corridor	0.08	binary
Valid extreme	0.14	binary

What lowers the score

Market-consistency reduction · up to 0.15

`0.15 × (1 - clip(gap / 0.35))`. A small gap means unit value moved similarly to the market, so more is subtracted. Once the gap reaches 0.35, the reduction falls to zero.

Data-reliability reduction · up to 0.08

`0.08 × (6 - quality score) / 6`. A quality score of 6 subtracts zero. Lower usable scores subtract progressively more. Ineligible rows are never scored.

The weighted sum is auditable but not statistically fitted. Its weights express documented review priorities; they do not estimate causal importance or probability.

Candidate model implementations

Logistic regression baseline

- Median imputation with missing indicators.
- Standard scaling.
- Balanced class weights.
- Liblinear solver; maximum 2,000 iterations.

A linear benchmark for whether a simple learned combination is sufficient.

XGBoost supervised challenger

- 160 trees; histogram tree method.
- Depth, learning rate, child weight, and L2 penalty compared on validation AP.
- Subsample and column sample 0.9; L1 penalty 0.15.
- Class imbalance handled with `scale_pos_weight`.

Captures nonlinear interactions between the 22 input fields.

Isolation Forest side signal

- Median imputation.
- 160 trees; fixed seed.
- Negative decision function min-max scaled to 0-1.

Unsupervised outlier reference, not selected for the official queue.

Weighted-sum baseline

- No fitting.
- Fixed weights and references in YAML.
- Component-level decomposition.
- Market and quality reductions.

Provides an inspectable review logic and the transparent component of the hybrid.

Selected XGBoost configuration

`max_depth=2`, `learning_rate=0.06`, `min_child_weight=2`, and `reg_lambda=4` produced validation average precision `0.3860`, the best of three deliberately bounded configurations. The seed is `20260117`; training uses 2017-2020 only.

Design trade-off. The search is intentionally narrow. It reduces tuning flexibility and overfitting opportunities, but it does not prove the chosen hyperparameters are globally optimal.

9 · SCORING IMPLEMENTATION

Hybrid selection and test discipline

The supervised challenger is chosen on 2021-2022 validation average precision. XGBoost wins. Three predefined blends then compete on validation precision@50.

$$P@k = \frac{TP_k}{k}$$

$$R@k = \frac{TP_k}{N_+}$$

XGBoost weight	Weighted-sum weight	Validation precision@50	Recall@50	AP	Hard-negative FPR
0.75	**0.25**	**0.46**	**0.213**	**0.309**	0.0139
0.50	0.50	0.32	0.148	0.264	0.0139
0.25	0.75	0.26	0.120	0.199	0.0139

FIT
2017-2020
Logistic, XGBoost, and Isolation Forest learn from the scenario test copy.

→

SELECT
2021-2022
Choose challenger, then predefined blend weight.

→

TEST ONCE
2023-2024
Measure the frozen design on unseen years.

→

APPLY
Official rows
Selected method scores real eligible observations only.

Standalone XGBoost later scores slightly higher than the hybrid on held-out test precision@50, `0.50` versus `0.48`. The hybrid remains selected because changing the method after seeing test results would turn the test into another selection set.

Metrics and held-out results

Precision@k

$$P@k = \frac{TP_k}{k}$$

Share of the top queue that contains planted positive patterns.

Recall@k

$$R@k = \frac{TP_k}{N_+}$$

Share of all planted positives recovered in the top queue.

Lift@k

$$L@k = \frac{P@k}{N_+/N}$$

Concentration relative to random selection from the evaluated split.

Average precision summarises ranking quality across thresholds. Hard-negative false-positive rate is the share of planted benign look-alikes that enter the Top 50. Ordinary false-positive rate uses all remaining negative rows and is reported as a diagnostic, not as a real-world false-alert rate.

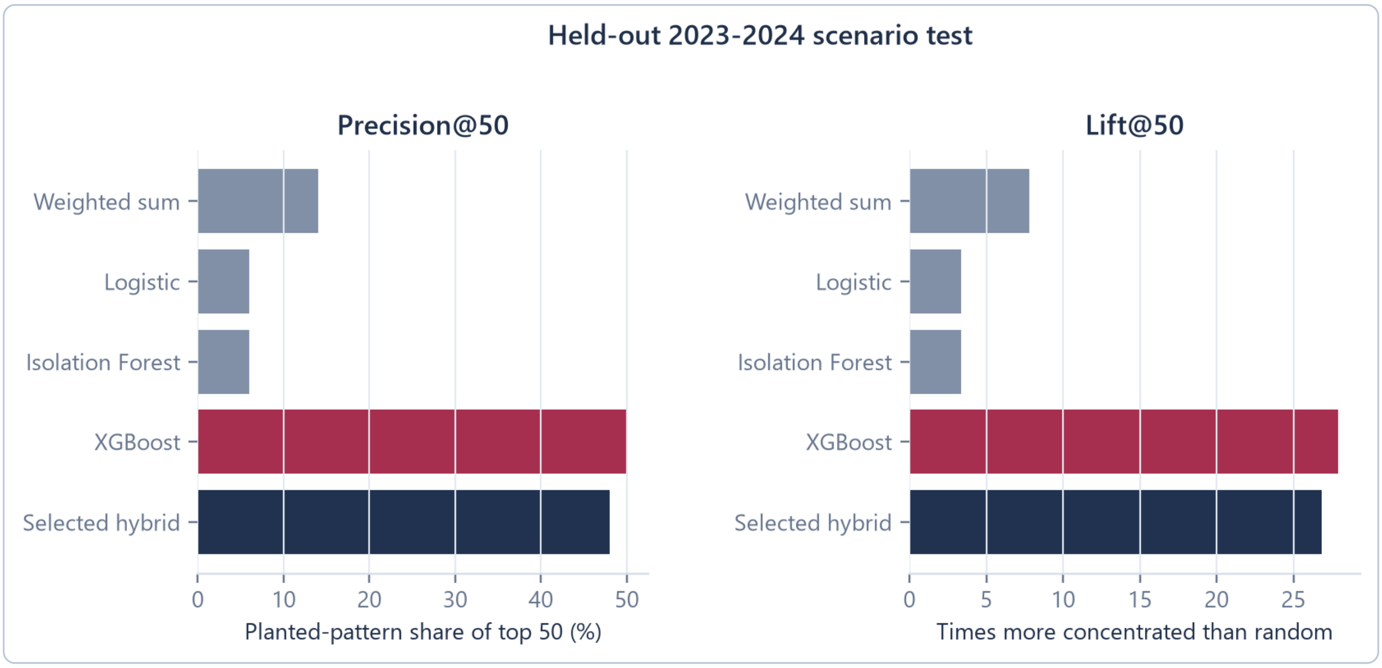

Figure 7. Held-out 2023-2024 controlled-scenario results. These are scenario-test metrics, not crime-detection rates.

The selected hybrid places 24 planted positives in its Top 50: precision `0.48`, recall `0.2222`, lift `26.81`, average precision `0.29785`, and hard-negative false-positive rate `0.0`.

Family variation and model interpretation

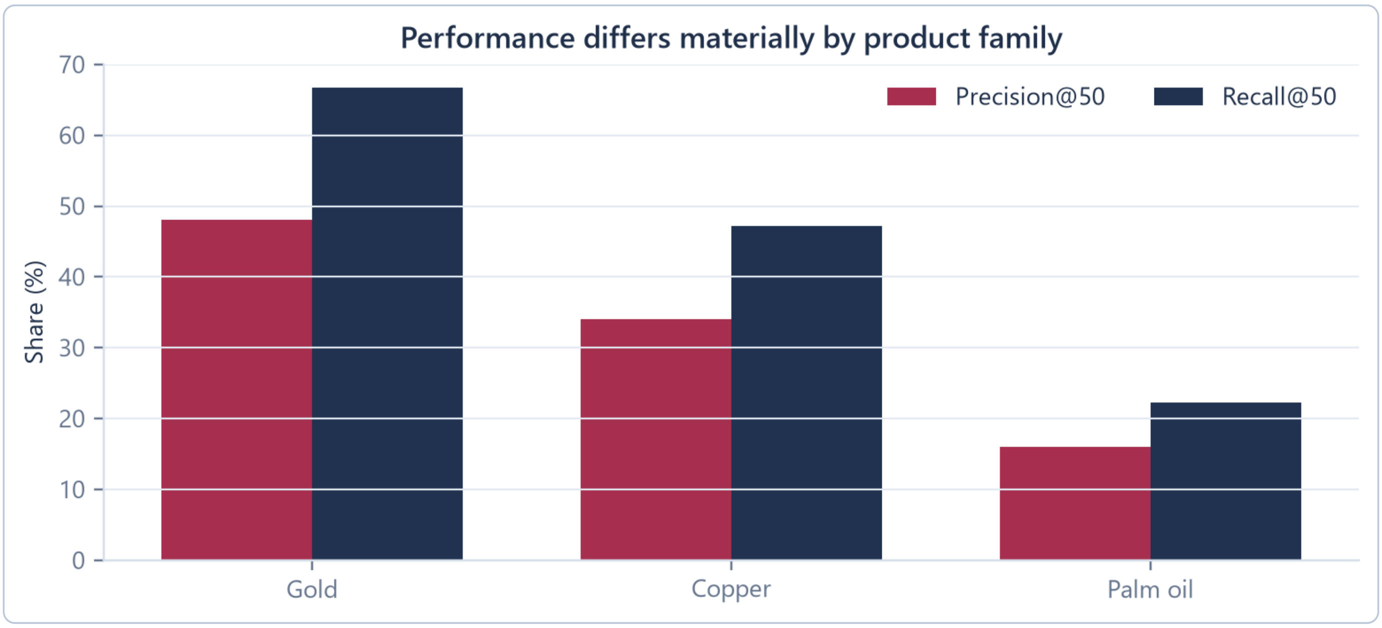

Figure 8. Family-level Top-50 metrics use a separate Top 50 within each product family; they are diagnostic and do not describe the mixed official queue.

The hybrid's held-out family results differ: gold precision@50 is `0.48` with recall `0.667`; copper is `0.34` / `0.472`; palm oil is `0.16` / `0.222`. The comparison shows heterogeneity, but each family has only 36 planted positives per split. It is evidence for monitoring family behaviour, not for declaring one commodity intrinsically easier or riskier.

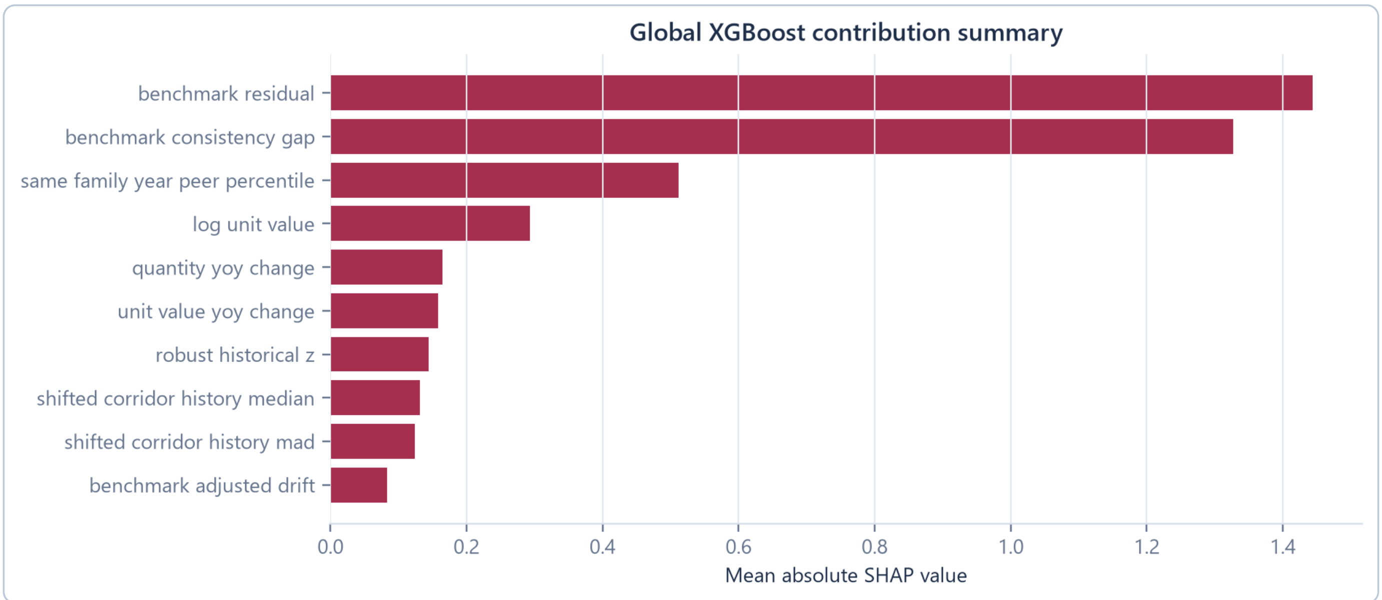

Figure 9. Mean absolute SHAP values for the XGBoost challenger. Magnitude shows average model contribution, not direction, causality, or case evidence.

Benchmark residual and benchmark-consistency gap dominate the global XGBoost summary, followed by same-family-year peer percentile and log unit value. This explains what the learned challenger uses most across scenario rows.

SHAP boundary. The project produces global SHAP only. Selected-case evidence comes from deterministic feature calculations and thresholds, not row-level SHAP.

Scoring official observations and constructing the queue

After selection, the fitted models and fixed weighted sum return to `corridor_features.parquet`. Only rows with valid unit value and an eligible quality status are scored.

ELIGIBILITY
23,656 official rows
Positive value and quantity; usable status.

→

SCORING
Frozen 75/25 hybrid
No scenario IDs or labels are present.

→

ORDERING
Score descending
`obs_id` is a deterministic tie-breaker.

→

PERSISTENCE
Top 50 CSV
Stage 07 adds evidence counts.

Rank		Year	Corridor	Product	Trade value (USD)	Quantity (metric tons)	Unit value (USD/t)	Benchmark price (USD/t)	Review-priority score
1	☑	2022	ESP → NLD	Gold	11,787,415	0.032	$368,356,718.75	$57,890,634.27	0.989
2	☐	2022	KWT → NPL	Gold	2,912,170	0.009	$323,574,444.44	$57,890,634.27	0.986
3	☐	2022	QAT → NPL	Gold	90,853,064	0.28	$324,475,228.57	$57,890,634.27	0.985
4	☐	2022	HKG → NPL	Gold	3,899,093	0.013	$299,930,230.77	$57,890,634.27	0.985
5	☐	2022	SAU → NPL	Gold	23,978,692	0.066	$363,313,515.15	$57,890,634.27	0.985
6	☐	2022	ITA → MKD	Palm oil	39	0.006	$6,500.00	$1,275.99	0.980
7	☐	2024	GBR → ITA	Copper	2,086	0.044	$47,409.09	$9,142.14	0.980
8	☐	2022	CAF → CHE	Gold	3,179,015	0.018	$176,611,944.44	$57,890,634.27	0.979
9	☐	2018	USA → IRL	Copper	264,595	6.879	$38,464.17	$6,529.80	0.979
10	☐	2020	ITA → TUR	Copper	24,289,083	628.199	$38,664.63	$6,173.77	0.975
11	☐	2022	SGP → JPN	Copper	112,739	2.09	$53,942.11	$8,822.37	0.975
[illegible]	☐	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Figure 10. Current live queue capture. Filters re-slice loaded rows; the CSV export retains full precision and rank order.

The dashboard exposes rank, year, corridor, product, trade value, quantity, implied unit value, annual benchmark price, and review-priority score. Exporter and importer filters display country names while preserving ISO3 values.

Rank reflects analytical ordering within this product and period scope. It is not a country-risk judgement, customer rating, or probability.

Evidence construction and selected-case analytics

Stage 07 converts the Top 50 into 200 evidence rows, exactly four per case. Candidate evidence types include history deviation, annual unit-value change, benchmark-adjusted drift, benchmark gap, and value-quantity divergence. The four strongest qualifying checks are retained.

Each evidence row carries:

- `evidence_id`, `obs_id`, source table, and source row ID;
- metric name, observed value, comparison value, group, and threshold;
- direction and severity;
- source fields and source version;
- plain-English summary and caveat.

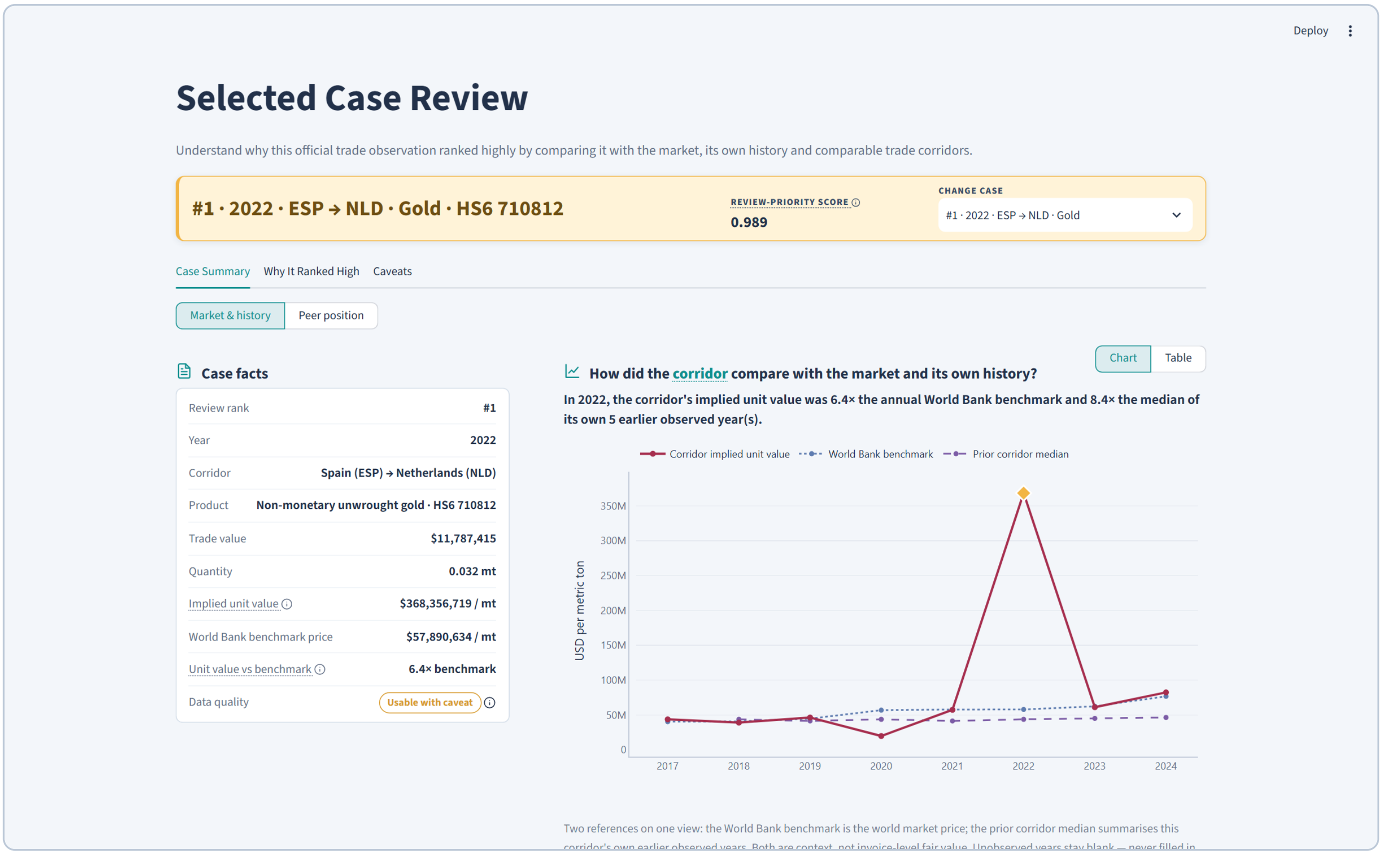

Figure 11. Current selected-case capture for rank 1: Spain to Netherlands, non-monetary unwrought gold, 2022.

The case page combines facts, market and corridor history, same-year peers, actual feature values, evidence cards, and limitations. Runtime checks reconcile chart and table values before display. Five deterministic analyst briefs are built from evidence IDs and pass citation, caveat, source, prohibited-language, and conclusion-boundary gates.

> **Design decision: deterministic evidence.** Model attribution can help reviewers understand a model globally. Case justification instead uses recomputable fields and explicit comparison rules.

Streamlit dashboard implementation

ENTRY	PAGES	COMPONENTS	SERVICES
`streamlit_app.py`	`app_pages/`	`components/`	`services/`
Page config, hidden router, custom navigation, boundary footer.	Nine stakeholder pages plus the welcome route.	Cards, charts, tables, filters, icons, styles, banners.	Loaders, metrics, contracts, state, paths, formatting.

Runtime design

Concern	Implementation
Paths	Repository root discovered using `configs/project.yml` and `src/tbml_common.py` markers
Data loading	Central `DATA_FILES` registry with required/optional status
Caching	`st.cache_data` on file and configuration loaders
Contracts	Required columns and integrity rules in `data_contracts.py`
Selection	`tbml_selected_obs_id` stored through typed session-state helpers
Filters	Re-slice cached dataframes; do not rerun the pipeline
Exports	Current queue view exported to CSV without modifying source files
Missing outputs	Required files raise a named regeneration error; optional panels show transparent empty states
Content	Stakeholder wording centralised in `dashboard_content.yml`
Theme	Project tokens in `dashboard_theme.yml`; Streamlit base theme in `.streamlit/config.toml`

The application is Python/Streamlit throughout. There is no React, Flask, database, API service, authentication layer, or LLM call in the current dashboard.

14 · DEPLOYMENT

GitHub and Streamlit Community Cloud

01
Local development
Pipeline, tests, dashboard, and document builds run in the repository environment.

→

02
Validated source and outputs
Generated public-data artefacts required by the dashboard are committed.

→

03
Git commit and push
Versioned code, configuration, and dashboard data move to GitHub.

04
GitHub repository
Streamlit Community Cloud reads the configured branch and entry point.

→

05
Cloud rebuild
`requirements.txt` installs the slim runtime; the app starts.

→

06
Live read-only dashboard
tbml-review-triage.streamlit.app

Verified deployment elements

- Entry point: `dashboard/streamlit_app.py`.
- Runtime dependencies: pinned Streamlit, Plotly, pandas, NumPy, PyArrow, and PyYAML in `requirements.txt`.
- Full pipeline dependencies: separate `requirements-pipeline.txt`.
- Streamlit theme/server settings: `.streamlit/config.toml`.
- Repository default working branch for this build: `develop`.
- No active GitHub Actions workflow is present for deployment.

The repository does not implement scheduled source refresh, automated retraining, production storage, API scoring, secrets, authentication, or operational monitoring. Streamlit Community Cloud serves committed public-data outputs.

> **Design trade-off: reproducible showcase versus live service.** Committed artefacts make the public project easy to inspect and deploy. They do not provide automated data freshness or production controls.

Validation, testing, and reproducibility

Risk	Control	Implementation	Current evidence
Wrong source or scope	Hash, schema, year, HS6, and count checks	stage 01	source manifest passes
Row loss / duplicate grain	Count reconciliation and canonical-key uniqueness	panel builder	25,844 retained
Unit error	Explicit value and gold benchmark conversions	common module/tests	current outputs reconcile
Time leakage	Shifted history; temporal splits	feature/scenario code	tests and manifest
Label leakage	Separate label table; forbidden fields rejected	stages 04-06	manifest and tests
Scenario contamination	Official queue rescored from official features only	stage 06	official-only queue test
Unsupported case claims	Evidence IDs, sources, caveats, language gates	stages 07-08	5/5 briefs pass
Dashboard schema drift	Data contracts and missing-output behaviour	dashboard services	page/data tests
Display inconsistency	Chart/table runtime gates	case services/tests	test suite passes

Current validation status

`157` tests pass across dashboard pages, services, metrics, source cards, theme contracts, model evaluation, trade landscape, selected-case calculations, and gold coverage. Pytest emitted one non-analytical warning because its cache directory was not writable in the document-build sandbox.

Core reproduction commands

```
python -m pip install -r requirements-pipeline.txt
python src/00_check_environment.py
python src/01_verify_sources.py
python src/02_ingest_official_sources.py
python src/03_build_panel.py
python src/04_build_features.py
python src/05_build_scenarios.py
python src/06_train_evaluate_models.py
python src/07_build_evidence.py
python src/08_build_briefs.py
python src/09_generate_reports.py
python -m pytest tests -q
python -m streamlit run dashboard/streamlit_app.py
```

Limitations and future bank pathway

Current limitations

The analysis is annual and aggregate. It does not observe invoices, shipments, customers, payment chains, counterparties, ownership, grade, purity, Incoterms, freight, or contract timing. World Bank series introduce basis risk; short histories can destabilise robust comparisons; missing quantity removes rows from unit-value scoring; scenarios test designed behaviours rather than confirmed outcomes; scores are uncalibrated; global SHAP is not case evidence; and committed outputs do not refresh automatically.

Future bank integration

Current implementation

External public-data signal → ranked official queue → recomputable evidence → human review starting point.

+

Future-state design only

Add governed KYC/CDD, trade documents, shipment/customs, payments, screening, ownership, case management, and disposition feedback.

An optional AI assistant could help locate related records, compare fields, organise a timeline, and cite agreements, conflicts, or gaps. It must not decide suspicion, disposition, or action. Every result would require source links, entitlements, audit logs, validation, privacy controls, and accountable human review.

Adoption path. Start with offline enrichment and no automated decisions; then controlled case-management integration and disposition capture; only then consider validated deployment, monitoring, and periodic recalibration.

This output prioritises an unusual corridor-product pattern for human review. It does not establish money laundering, misinvoicing, or criminal intent.

Feature reference: levels, history, and peers

Feature	Definition / formula	Grain and time rule	Missing / edge handling	Primary use
`log_unit_value`	$\log(V_t/V_{t^-})$	current corridor-product-year	missing if unit value invalid	ML; basis for history
`shifted_corridor_history_median`	median prior log unit value	same corridor+HS6, years before current	missing at first usable history	ML; history evidence
`shifted_corridor_history_mad`	median absolute deviation around shifted median	prior-only	missing without prior history	ML; robust z denominator
`robust_historical_z`	$\log(Q_t/Q_{t^-})$	prior-only	missing if median missing; unscaled MAD	rules; ML; evidence
`same_family_year_peer_percentile`	percentile rank of benchmark residual	same family and current year	average rank for ties	rules; ML; peer view
`benchmark_residual`	$\log(UV_t/UV_{t^-})$	current row and current-year benchmark	missing if unit value/benchmark absent	rules; ML; evidence
`benchmark_price_usd_per_metric_ton`	annual World Bank benchmark, normalised to USD/mt	family-year	matched for all scoped rows	context/dashboard
`benchmark_yoy_change`	log benchmark change from prior year	family-year	missing in first year	ML; consistency gap
`data_quality_score`	quantity 1 + benchmark 2 + history 0-2 + completeness 1	current row plus prior-count depth	0-6	ML; weighted-sum reduction
`valid_extreme_flag`	unit value >5× or <0.2× benchmark after validity checks	current row	false if ratio unavailable	rules; caveat

The model input list is defined once in `feature_columns()`. Identity fields such as `obs_id`, year, country codes, corridor ID, HS6, family, and product name travel alongside the features but are not model inputs.

Feature reference: changes, status, and missingness

Feature	Definition / formula	Grain and time rule	Missing / edge handling	Primary use
`trade_value_yoy_change`	$\log(V_t/V_{t^-})$	previous observed same corridor+HS6 row	missing without valid prior/ current values	ML
`quantity_yoy_change`	$\log(Q_t/Q_{t^-})$	previous observed row	missing without valid quantity pair	ML
`unit_value_yoy_change`	$\log(UV_t/UV_{t^-})$	previous observed row	missing without valid unit values	rules; ML; evidence
`benchmark_adjusted_drift`	$r_t - r_{t^-}$	current and prior residual	missing without prior residual	rules; ML; evidence
`value_quantity_divergence`	$\Delta\log V - \Delta\log Q$	prior observed row	missing if either change missing	rules; ML; evidence
`benchmark_consistency_gap`	$\lvert\Delta\log UV - \Delta\log B\rvert$	row movement vs current family benchmark movement	missing when change unavailable	weighted-sum reduction; ML
`corridor_activity_history`	number of earlier same corridor+HS6 rows	prior-only cumulative count	zero at first appearance	ML
`corridor_novelty_flag`	1 when prior activity count is zero	prior-only	binary	rules; ML
`corridor_reactivation_flag`	1 when route returns after >1 calendar-year gap	prior observed year	binary	rules; ML
`missing_quantity_flag`	quantity is null	current row	binary	eligibility; ML
`missing_benchmark_flag`	benchmark join is not matched	current row	binary	ML/audit
`missing_history_flag`	shifted history median is null	prior-only	binary	ML/audit
`missing_country_mapping_flag`	exporter or importer ISO3 missing	current row	binary	ML/audit

`trade_value_yoy_change` is named "year-over-year" in outputs, but technically compares with the previous observed row in the group. If a corridor disappears for a year, the gap is captured separately by `corridor_reactivation_flag`.

Key data and output fields

Field	Table	Meaning	Unit / type	Caveat
`obs_id`	panel/features/queue/evidence	stable analytical observation ID	string	not a source-system transaction ID
`source_row_id`	panel/evidence	stable link to filtered BACI row	string	filtered-extract lineage
`year`	all row tables	calendar year	integer	annual aggregation
`corridor_id`	panel/features	exporter-importer identifier	string	direction matters
`hs6`	panel/features/queue	six-character product code	string	one exact code per family
`trade_value_usd`	panel/features/queue	BACI value × 1,000	current USD	annual aggregate
`quantity_metric_ton`	panel/features/queue	reported BACI quantity	metric tons	may be unavailable
`unit_value_usd_per_metric_ton`	panel/features/queue	value divided by quantity	USD/mt	implied average, not invoice price
`quality_status`	panel/features/queue	usability classification	category	caveat is not suspicion
`model_eligible`	panel/features	valid unit value and usable status	boolean	excludes 2,188 rows
`selected_review_priority_score`	queue	frozen hybrid ranking signal	0-1 score	not calibrated probability
`rank`	queue	deterministic descending score order	integer	scoped to current queue
`evidence_id`	evidence	stable evidence-row ID	string	one case has four current checks
`observed_value`	evidence	metric value behind evidence	numeric	interpretation depends on metric
`comparison_value`	evidence	benchmark, prior, or peer reference	numeric	may be absent for change metrics
`caveat`	evidence	required interpretation boundary	text	accompanies the evidence fact

Full generated field definitions remain in `reports/data_dictionary.md`; the live dashboard presents a curated stakeholder subset.

Model and selection configuration

Configuration	Current value
Primary seed	`20260117`
Train / validation / test	2017-2020 / 2021-2022 / 2023-2024
Model inputs	22 fields from `feature_columns()`
Logistic	median imputer + indicators; standard scaler; balanced liblinear
XGBoost trees	160
XGBoost selected	depth 2; learning rate 0.06; child weight 2; L2 4
XGBoost fixed	subsample 0.9; column sample 0.9; L1 0.15; histogram tree method
Isolation Forest	160 estimators; median imputer; fixed seed
Challenger selection	highest validation average precision among logistic and XGBoost
Blend candidates	challenger weights 0.25, 0.50, 0.75
Blend selection	validation precision@50; then hard-negative FPR; then AP; then lower challenger weight
Selected score	0.75 XGBoost + 0.25 weighted sum
Official persistence	Top 50

Logistic coefficient interpretation

Coefficients apply after median imputation, added missing indicators, and standard scaling. Their sign and magnitude describe the fitted linear scenario model, not causal effects. The largest absolute current coefficients include benchmark-consistency gap, corridor activity history, benchmark-adjusted drift, shifted-history MAD, and unit-value movement.

Stored model bundle

`model_bundle.joblib` contains the feature list, fitted logistic pipeline, fitted XGBoost model, fitted Isolation Forest, and selection record. Reusing it avoids retraining when the same prepared features are rescored.

Controlled scenario catalogue

Scenario ID	Label	Per split	Transformation	Intended test
`synthetic_overvaluation`	positive	36	value × uniform 2.2-3.5; quantity fixed	recover sharp implied-value rise
`synthetic_undervaluation`	positive	36	value × uniform 0.25-0.45; quantity fixed	recover sharp implied-value fall
`synthetic_low_volume_high_value`	positive	36	value × 1.8-2.8; quantity × 0.35-0.65	recover value/quantity divergence
`hard_negative_benchmark_consistent_movement`	negative	36	align unit value to family-year median residual	avoid market-consistent over-alert
`hard_negative_proportional_value_quantity_growth`	negative	36	multiply value and quantity by same factor	avoid flagging scale growth with stable unit value

Split manifest

Split	Years	Panel rows	Positive scenarios	Hard negatives
Train	2017-2020	12,318	108	72
Validation	2021-2022	6,868	108	72
Test	2023-2024	6,658	108	72

Scenario sampling requires model eligibility, positive quantity, a matched benchmark, and enough rows in each family-year pool. A single configured seed drives the current injection manifest. Scenario labels remain physically separated from features.

Data sources and provenance

Publisher / dataset	Release / period	Project file	Native unit	Project role	Key limitation
CEPII BACI HS17	V202601; 2017-2024	`baci_hs17_v202601_selected_...parquet`	value in thousand USD; quantity in metric tons	official-derived bilateral trade facts	annual aggregates; reporting and reconciliation limits
CEPII country codes	V202601	`country_codes_V202601.csv`	code/name	exporter/importer mapping	mapping reference only
CEPII HS17 products	V202601	`product_codes_HS17_V202601.csv`	six-character HS6	product description	product code cannot reveal grade or contract
World Bank Commodity Markets	annual; 2017-2024	`CMO-Historical-Data-Annual.xlsx`	USD/mt; gold USD/troy oz	broad market benchmark	not corridor-specific fair value
FATF-Egmont TBML trends	2020	`Trade-Based-Money-Laundering-Trends-and-Developments_2020.pdf`	publication	typology and caveat context	never a model input or case fact
FATF-Egmont risk indicators	2021	`Trade-Based-Money-Laundering-Risk-Indicators_2021.pdf`	publication	review-question context	indicators are not labels

Official links:

- CEPII BACI database
- World Bank Commodity Markets
- FATF-Egmont TBML Trends and Developments
- FATF-Egmont TBML Risk Indicators

The source manifest records SHA-256 hashes, row counts, year and HS6 scope, workbook sheets, and reference-file inventory.

Reproducibility and deployment reference

Repository structure

Path	Responsibility
`configs/`	project scope, products, benchmarks, thresholds, scenarios, typology
`src/`	offline pipeline stages and shared analytical functions
`data/raw`, `interim`, `processed`, `outputs`	source, staged, analytical, and presentation artefacts
`reports/`	generated analysis, model card, dictionaries, figures, companion
`dashboard/`	Streamlit pages, components, services, and content/theme config
`tests/`	analytical and dashboard validation
`.streamlit/config.toml`	deployed Streamlit base theme and server settings

Environment split

`requirements-pipeline.txt` supports data ingestion, modelling, notebooks, reporting, and tests. The deployed `requirements.txt` contains only the packages imported by the read-only dashboard. This reduces cloud build time and attack surface, while requiring pipeline artefacts to be generated before deployment.

Launch and document build

```
python -m pytest tests -q
python -m streamlit run dashboard/streamlit_app.py
python reports/technical_companion/build_technical_companion.py
```

Operational limitations

No automated refresh schedule, retraining job, database, API, authentication, secrets integration, or monitoring service is implemented. Reproducibility depends on the recorded source release, hashes, configuration, environment, and Git commit.

Glossary and implementation traceability

Glossary

Term	Meaning
AFC	Anti-financial crime
TBML	Trade-based money laundering
HS6	Six-digit Harmonized System product code
Corridor	Directional exporter-importer pair
Unit value	Aggregate value divided by aggregate quantity
MAD	Median absolute deviation
Time leakage	Future information entering an earlier assessment
Hard negative	Controlled benign pattern designed to look unusual
Lift@k	Precision@k divided by positive prevalence
SHAP	Model-output contribution method; not case evidence

Dashboard traceability

Dashboard page	Source artefact	Calculation / module
Business Problem and Value	panel, project config	`dashboard_metrics.py`
From Data to Review Queue	manifests, panel, features	page composition + shared services
Trade Landscape and Patterns	panel, queue, benchmarks	`dashboard_metrics.py`, charts
Top 50 Review Queue	queue + features	filters, tables, session state
Selected Case Review	panel, features, evidence	`case_summary.py`, `why_ranked_high.py`
Model Evaluation & Controls	model scores/comparison/config	model page + integrity report
Unscored Gold Records	panel	`gold_coverage.py`
Bank Implementation Pathway	content config	future-state presentation only
Data Dictionary	reports and curated metadata	`data_dictionary.py`

Final interpretation boundary
This project demonstrates disciplined review prioritisation over public annual trade data. Investigation, disposition, and any conclusion about wrongdoing remain human responsibilities supported by evidence the current dataset does not contain.

Build identity: commit `774aac6` · BACI V202601 · analysis 2017-2024 · generated 23 July 2026 · live dashboard · repository